# Arafat Rahman Oany

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Arafat Rahman Oany  Arafat Rahman Oany


## Education

***Doctor of Phylosophy***
Biomedical Science
(2023-Present)

***Biomedical Science (BIMS) PhD Rotation Program***
*Department of Veterinary Physiology & Pharmacology*
*Texas A&M University, College Station, Texas 77843, United States*
***Advisor:*** *Professor Stephen Safe*
***Research Focus:*** *Ferroptosis and its therapeutic potential in cancer.*

***Master of Science***
Biotechnology and Genetic Engineering
(2014-2016)

***Thesis:*** Isolation, identification and molecular characterization of *Shigella flexneri* specific bacteriophages from the environmental waters in Bangladesh
***Advisor:*** Dr. Kaiser Ali Talukder (Former PI, Enteric and Food Microbiology Laboratory, **icddr,b**)
Mawlana Bhashani Science and Technology University, Bangladesh.

***Bachelor of Science***
Biotechnology and Genetic Engineering
(2010-2014)

***Research report***: A Hypothetical Protein of *Alteromonas macleodii* AltDE1 (amad1_06475) Predicted to be a Cold-Shock Protein with RNA Chaperone Activity
***Advisor:*** Dr. Shah Adil Ishtiyaq Ahmad
Mawlana Bhashani Science and Technology University, Bangladesh

## Research Interest

Cancer biology, Drug designing & Computational biology.

## Research Experience

**Jan 24 to present** — **Graduate Research Assistant**
**Supervised By:** **Professor Stephen Safe**
Molecular & Cellular Oncology Laboratory (Safe Lab)
Department of Veterinary Physiology & Pharmacology
*Texas A&M University, College Station, Texas 77843, United States*
**Project:** Cancer drug development.
**Responsibilities:** Research design, conduct different experiments, data acquisition and analysis.

**Aug 23 to Dec 23** — **PhD Lab Rotation**
**Supervised By:** **Professor Sanjay Reddy**
Department of Veterinary Pathobiology
*Texas A&M University, College Station, Texas 77843, United States*
**Project:** Vaccine development for Marek's disease virus.
**Responsibilities:** Research design, conduct different experiments, data acquisition and analysis.

**Oct 2015 to Nov 16** — **Elective Research fellow** (National Field Experience)
**Supervised By:** Dr. Kaiser Ali Talukder
Enteric and Food Microbiology Laboratory
International Centre for Diarrheal Disease Research, Bangladesh (**icddr, b**)
**Project:** Serotype specific bacteriophage isolation
**Responsibilities:** Research design, sample collection, conduct routine tests, data preparation and analysis.
**Goal:** Serotype specific bacteriophage isolation, characterization, and evaluation for alternative treatment of diarrhea

## Publications (Peer-reviewed)

**Google Scholar ID:** Arafat Rahman Oany  **ORCID:** 0000-0003-2228-457X  NCBI Bibliography
**Total Citation:** **447**
- **h-Index: 13**
- **i10-index: 13**

***Manuscript published:***

1. **Oany AR**, Srijana Upadhyay, Stephen Safe. Dual NR4A1/2 ligands inhibit rhabdomyosarcoma cell growth and trigger ferroptosis by activating CD71 [abstract]. In: Proceedings of the AACR-NCI-EORTC International Conference on Molecular Targets and Cancer Therapeutics; 2025 Oct 22-26; Boston, MA. Philadelphia (PA): AACR; Mol Cancer Ther 2025;24(10 Suppl): Abstract nr C074.

https://doi.org/10.1158/1535-7163.TARG-25-C074

2. **Oany, A.R.,** Upadhyay, S., Tsui, W.N., Hailemariam, A., Latka, S., Landua, J., Scherer, S., Welm, A.L., Villanueva, H. and Lewis, M., Safe, S., 2025. Orphan Nuclear Receptor 4A1 (NR4A1) and NR4A2 are Endogenous Regulators of CD71 and Their Ligands Induce Ferroptosis in Breast Cancer. (***Cell Death & Disease***). https://pubmed.ncbi.nlm.nih.gov/40313760/
3. Tsui, W.N.T., Park, Y., Upadhyay, S., Kim, D.M., Zhang, L., Wright, G., Hailemariam, A., Oany, A.R., Han, S.J. and Safe, S., 2025. Dual Targeting of Orphan Nuclear Receptors NR4A1 and NR4A2 for Nonhormonal Endometriosis Therapy. *Endocrinology*, 166(11), p.bqaf144. (Impact Factor: **3.3**) https://doi.org/10.1210/endocr/bqaf144
4. Safe, S., **Oany, A.R.**, Tsui, W.N., Lee, M., Srivastava, V., Upadhyay, S., Hailemariam, A., Farkas, E., Kakwan, S., Kearns, C. and Sivaram, G., 2025. Orphan nuclear receptor transcription factors as drug targets. Transcription, 16(2-3), pp.224-260. (Impact Factor: **4.4**) https://doi.org/10.1080/21541264.2025.2521766
5. Hailemariam, A., Upadhyay, S., Srivastava, V., Hafiz, Z., Zhang, L., Tsui, W.N.T., **Oany, A.R.**, Rivera-Rodriguez, J., Chapkin, R.S., Riddell, N. and McCrindle, R., Perfluorooctane Sulfonate (PFOS) and Related Compounds Induce Nuclear Receptor 4A1 (NR4A1)-Dependent Carcinogenesis. Chemical research in toxicology. (Impact Factor: **3.7**) https://doi.org/10.1021/acs.chemrestox.4c00528
6. Safe, S., Farkas, E., Hailemariam, A.E., **Oany, A.R.**, Sivaram, G. and Tsui, W.N.T., 2025. Activation of Genes by Nuclear Receptor/Specificity Protein (Sp) Interactions in Cancer. *Cancers*, 17(2), p.284. (Impact Factor: **4.5**) https://doi.org/10.3390/cancers17020284
7. Upadhyay, S., Lee, M., Zhang, L., **Oany, A.R.**, Mikheeva, S.A., Mikheev, A.M., Rostomily, R.C. and Safe, S., 2024. Dual nuclear receptor 4a1 (nr4a1/nr4a2) ligands inhibit glioblastoma growth and target TWIST1. *Molecular Pharmacology*, 107 (02) p.100009. (Impact Factor: **3.2**) https://doi.org/10.1016/j.molpha.2024.100009
8. Mia, M.; **Oany AR.**; Pervin, T., 2024. Deciphering novel molecular gene expression signatures and pathways in cystic fibrosis through integrative bioinformatics strategies. Gene & Protein in Disease, 3(2), 2937. https://doi.org/10.36922/gpd.2937
9. **Oany AR.**; Mia, M.; Pervin, T.; Alyami, S.A.; Moni, M.A 2021. Integrative Systems Biology Approaches to Identify Potential Biomarkers and Pathways of Cervical Cancer. *Journal of Personalized. Medicine.*, 11, 363. (PMID: 33946372) (Impact Factor: **3.0**) https://doi.org/10.3390/jpm11050363.
10. **Oany AR.**, Pervin, T. and Moni, M.A., 2021. Pharmacoinformatics-based elucidation and designing of potential inhibitors against Plasmodium falciparum to target importin α/β mediated nuclear importation. *Infection, Genetics and Evolution*, 88, p.104699. (PMID: 33385575) (Impact Factor: **2.6**) https://doi.org/10.1016/j.meegid.2020.104699.
11. Pervin, T. and **Oany AR.**, 2021. Vaccinomics approach for scheming potential epitope-based peptide vaccine by targeting l-protein of Marburg virus. *In silico* pharmacology, 9(1), pp.1-18. (PMID: 33717824) https://doi.org/10.1007/s40203-021-00080-3.
12. **Oany AR.**, Mia, M., Pervin, T., Junaid, M., Hosen, S.Z. and Moni, M.A., 2020. Design of novel viral attachment inhibitors of the spike glycoprotein (S) of severe acute respiratory syndrome coronavirus-2 (SARS-CoV-2) through virtual screening and dynamics. *International journal of antimicrobial agents*, 56(6), p.106177. (PMID: 32987103) (Impact Factor: **4.9**) https://doi.org/10.1016/j.ijantimicag.2020.106177
13. **Oany AR**, Pervin T, Mia M, et. Al, 2017. Vaccinomics approach for designing potential peptide vaccine by targeting *Shigella* Spp. serine protease autotransporter (SPATE) subfamily protein, SigA. *Journal of Immunology Research*.14. (PMID: 25187696) (Impact Factor: **3.5**) https://doi.org/10.1155/2017/6412353.
14. Hossain MU, Omar TM, **Oany AR**, Kibria KK, Shibly AZ, Moniruzzaman M, Ali SR, Islam MM, **2018**. Design of peptide-based epitope vaccine and further binding site scrutiny led to groundswell in drug discovery against Lassa virus. *3 Biotech*. Feb 1; 8(2):81. (PMID: 29430345) (Impact Factor: **2.6**) https://doi.org/10.1007/s13205-018-1106-5.
15. **Oany AR,** Emran AA, and Jyoti TP, 2014. Design of an epitope-based peptide vaccine against spike protein of human coronavirus: an *in silico* approach. *Drug design, development and therapy*. 8, 1139. (PMID: 25187696) (Impact Factor: **4.7**) (https://doi.org/10.2147/DDDT.S67861).
16. Kibria KK, Hossain MU, **Oany AR**, Ahmad SA, 2016. Novel insights on ENTH domain-containing proteins in apicomplexan parasites. *Parasitology Research*. 115, 2191. (PMID: 26922178) (Impact Factor: **1.8**) (https://doi.org/10.1007/s00436-016-4961-1).
17. Hossain MU, **Oany AR**, Ahmad SA, Hasan MA, Khan MA, Siddaki MA, 2016. Identification of potential inhibitor and enzyme-inhibitor complex on trypanothione reductase to control Chagas disease. *Computational Biology and Chemistry*: 65, 29. (PMID: 27744094) (Impact Factor: **2.6)** (https://doi.org/10.1016/j.compbiolchem.2016.10.002).
18. **Oany AR,** Hossain MU, Islam R, Emran *AA,* 2016. A preliminary evaluation of cytotoxicity, antihyperglycemic and antinociceptive activity of *Polygonum hydropiper* L. ethanolic leaf extracts. *Clinical Phytoscience.* 2, 1-6. (https://doi.org/10.1186/s40816-016-0016-5),
19. **Oany AR,** Sharmin T, Chowdhury AS, Jyoti TP, Hasan MA, 2015. Highly conserved regions in Ebola virus RNA

dependent RNA polymerase may act as a universal novel peptide vaccine target: a computational approach." *In Silico Pharmacology.*:3, 1-13. (PMID:26820892) (https://doi.org/10.1186/s40203-015-0011-4)

20. **Oany AR,** Jyoti TP, Ahmad SA, 2014. An *in silico* approach for characterization of an aminoglycoside antibiotic resistant methyltransferase protein from *Pyrococcus furiosus* (DSM 3638). *Bioinformatics and Biology Insights: 8* 65–72. (PMID: 24683305) (https://doi.org/10.4137/BBI.S146).
21. **Oany AR,** Ahmad SA, Kibria KK, Hossain MU, Jyoti TP, 2014. A hypothetical protein of *Alteromonas macleodii* AltDE1 (amad1_06475) predicted to be a cold-Shock protein with RNA chaperone activity. *Gene regulation and systems biology*:8 141. (PMID: 25574135)) (https://doi.org/10.4137/GRSB.S20802).
22. **Oany, A.R.**, Ahmed, M.S., Jahan, N., Latif, M.A., Mahmud, S., Hossain, M.A., Akter, F., Rakib, H.H. and Islam, M.S., 2015. Homology modeling and assigned functional annotation of an uncharacterized antitoxin protein from *Streptomyces xinghaiensis*. *Bioinformation*, 11(11), p.493. (PMID: 26912949). (https://doi.org/10.6026/97320630011493)
23. **Oany AR**, Ahmad SA, Hossain MU, Jyoti TP, 2015. Identification of highly conserved regions in L-segment of Crimean–Congo hemorrhagic fever virus and immunoinformatic prediction about potential novel vaccine. Advances and Applications in Bioinformatics and Chemistry: 8,1-10. (PMID: 25609983) (https://doi.org/10.2147/AABC.S75250)
24. **Oany, A.R.**, Hossain, M.U. and Ahmad, S.A.I., 2015. Computational approach to design a potential SiRNA molecule to silence the Nucleocapsid gene of different Nipah virus strains of Bangladesh. *Bioresearch Communications-(BRC)*, 1(1), pp.40-44.

## Conference/Symposium (Presentation)

### *Poster:*

1. *"Dual NR4A1/2 ligands inhibit rhabdomyosarcoma cell growth and trigger ferroptosis by activating CD71"*
   **Organized by**: AACR-NCI-EORTC International Conference on Molecular Targets and Cancer Therapeutics; 2025 Oct 22-26; Boston, MA.
2. *"Dual NR4A1/2 ligands inhibit breast cancer cell growth and trigger ferroptosis by activating CD71"*
   **Organized by**: Texas A&M Center for Environmental Health Research (TiCER), TX. (2025)
3. *"Dual NR4A1/2 ligands inhibit breast cancer cell growth and trigger ferroptosis by activating transferrin receptor (CD71)"*
   **Organized by**: *Cancer and Metabolism (abcam), Austin, TX (2025).*
4. *"Orphan Nuclear Receptor 4A1 (NR4A1) and NR4A2 are Endogenous Regulators of CD71 and Their Ligands Induce Ferroptosis in Breast Cancer "*
   ***Organized by***: Texas A&M University Regional Center of Excellence in Cancer Research (TREC) (2025).

## Awards and Honors:

1. Nominated by Texas A&M University to present at the Global Young Scientists Summit (GYSS) 2026 in Singapore.
2. American Association for Cancer Research (AACR)- Scholar-in-Training Award-($750) (2025).
3. Veterinary Medicine & Biomedical Sciences (VMBS)-Graduate Student Association (GSA) travel award (Fall 2025) ($1000) Texas A&M University, College Station, Texas, USA.
4. Veterinary Medicine & Biomedical Sciences (VMBS) Graduate Student Advanced Developmental Training Grant ($1250) (Spring 2025), Texas A&M University, College Station, Texas, USA.
5. Proteintech Product Grant-2025 ($1250).
6. Best poster award -Texas A&M Center for Environmental Research (TiCER) (Spring 2025), Texas A&M University, College Station, Texas, USA.
7. Travel award -Texas A&M Center for Environmental Research (TiCER) ($750) (Summer 2025), Texas A&M University, College Station, Texas, USA.
8. CIRTL Associate Fellow of the Academy for Future Faculty (AFF)- (Spring 2025), Texas A&M University, College Station, Texas, USA.
9. Judge certificate (Spring 2025)- in recognition of your contribution and dedication as a judge during Student Research Week 2025 at Texas A&M University, College Station, Texas, USA.
10. Veterinary Medicine & Biomedical Sciences (VMBS)-Graduate Student Association (GSA) travel award (Spring 2025) ($1000) Texas A&M University, College Station, Texas, USA.
11. Graduate student of the year award (2024), Department of Veterinary Physiology & Pharmacology, Texas A&M University, College Station, Texas, USA.
12. Texas A&M University Walter W. Lechner Estate Scholarship and have been designated a Lechner Scholar for the 2023-2024 academic year.
13. Bangladesh Sweden trust fund travel award for outstanding study abroad with scholarship (2023-24).
14. VMBS International photo contest (Category D. Experiencing Aggieland, 2024) first prize winner.
15. Community of the scholar engagement, Texas A&M University, Fall-24 (protégé).

## Technical sills

### *Laboratory skills*

- SDS-PAGE & Western blot
- Polymerase Chain Reaction (PCR)
- Real time PCR
- Cell Culture techniques
- Transfection & Transformation
- Cloning techniques
- Infrared Spectroscopy (ATR)
- UV –VIS Spectroscopy
- Gene silencing
- Atomic Absorption Spectroscopy (AAS)
- High Performance Liquid Chromatography (HPLC)
- Gas chromatography (GC)
- Thin Layer Chromatography (TLC)
- Pulse Field Gel Electrophoresis (PFGE)
- Restriction digestion and dendrogram
- ELISA
- Microbiological lab techniques

### *Analytical Tools*

- Data Analysis: GraphPad Prism.
- Bioinformatics: [BLAST, MSA, Docking, Primer designing, Protein structure modelling and Phylogenetic analysis, etc.].
- Good command of Microsoft Office™ tools.

### *Language skills*

- Mother tongue: Bengali
- Other language: Well skilled in English.

## Leadership & Service

***Ad Hoc Reviewer*: (My Web of Science Profile)**

- Scientific Reports: 16 reviews.
- Informatics in Medicine Unlocked: 3 reviews.
- Computers in Biology and Medicine: 3 reviews.
- BMC Immunology: 2 reviews.
- BMC Infectious Diseases: 2 reviews.
- BMC Bioinformatics: 2 reviews.
- BMC Cancer: 1 review and others.

***Student Mentoring*:**

1. Israt Jahan (Ph.D. student, Biochemistry & Molecular Biophysics, Texas A&M) (Spring-2025).
2. Prerna Tikute (Ph.D. student, Biomedical Science (BIMS), Texas A&M) (Spring-2025).

***Human Resource Officer*,** ISMA Texas A&M University — August 2025 to present.

## Training and Workshop

1. **Name**: Mouse Techniques
   ***Organized by***: Comparative Medicine Program, Division of Research, Texas A&M University, College Station, Texas, USA. **Date**: 18 September 2025.
2. **Name**: Three Minute Thesis Half-Day Workshop
   ***Organized by***: Graduate and Professional School, Texas A&M University, College Station, Texas, USA.
   **Date**: 26 September 2024.
3. **Name**: Mouse Handling/Restraint
   ***Organized by***: Comparative Medicine Program, Division of Research, Texas A&M University, College Station, Texas, USA. **Date**: 17 July 2024.
4. **Name**: Scientific Writing: A Survival Guide
   ***Organized by***: College of Veterinary Medicine & Biomedical Sciences, Texas A&M University, College Station, Texas, USA.
   **Date**: 17 July 2024.
5. **Name**: Biosafety and Biosecurity
   ***Organized by***: Biosafety division, ICDDR, B
   **Date**: 13-15 October 2015.

## Professional Experience

**Mar 2021 to Aug 2022 Senior Executive, Quality Control** (Aristopharma Ltd., Bangladesh)

***Responsibilities***:

• Led method validation processes, ensuring compliance with international regulatory standards (e.g., WHO GMP, ISO).

• Developed and updated Standard Operating Procedures (SOPs) to maintain consistency and

**Jan 2019 to Mar 2021 Senior Officer, Quality Control** (Aristopharma Ltd., Bangladesh)

***Responsibilities***:

•Conducted comprehensive quality analysis of raw materials, in-process products, and finished goods, adhering to pharmacopeial standards (USP, BP, EP).

**Nov 2016 to Jan 2019** **Quality Control Officer** (Aristopharma Ltd., Bangladesh)

***Responsibilities:***

• Performed routine and batch-wise quality control tests for raw materials of pharmaceutical products.

## Memberships

1. Student member, American Society of Clinical Oncology (ASCO) (ID: 7471159)
2. Associate member, American Association for Cancer Research (AACR ID:1393035)
3. General member, Graduate Student Association (GSA), VMBS, TAMU. (2023-Present)
4. Student member, Society of Toxicology (SOT) (2024-Present)
5. ASM Global Outreach member. (2014 - Present).
6. International Society for Computational Biology (ISCB) (2015-2020)
7. European Society of Clinical Microbiology and Infectious Diseases (ESCMID) (2015-2018)

## References

Dr. Stephen Safe
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